

# Replication Plan

Updated Virophage Taxonomy and Distinction from Polinton-like Viruses

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Auto-generated Replication Blueprint

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## Paper Overview

This paper provides an updated taxonomy of virophages—small viruses that parasitize giant viruses—and establishes criteria to distinguish them from related Polinton-like viruses (PLVs). The authors collect virophage and PLV genome sequences from public databases, identify conserved marker genes using HMM profiles, perform phylogenetic analyses (particularly of the major capsid protein (MCP) and packaging ATPase), analyze gene content and genome organization, and propose quantitative taxonomic thresholds (e.g., amino acid identity cutoffs) for genus/family-level classification.

## Detailed Workflow

### Phase 1: Genome Collection

#### 1.1 Download known virophage genomes from NCBI GenBank.

- Sputnik, Mavirus, Zamilon, and all others listed in the paper’s supplementary tables.

#### 1.2 Download Polinton-like virus genomes.

#### 1.3 Download giant virus genomes (Mimiviridae) for context.

#### 1.4 Retrieve metagenomic virophage sequences from IMG/VR or other databases.

### Phase 2: Gene Identification

#### 2.1 Predict ORFs in all genomes (Prodigal or GeneMarkS).

#### 2.2 Identify marker genes using HMM profiles:

- Major capsid protein (MCP).
- Packaging ATPase.
- Minor capsid protein, protease, integrase.

#### 2.3 Run HMMER (hmmsearch) against custom HMM profiles.

#### 2.4 Annotate all ORFs using BLAST against NR or InterProScan.

### Phase 3: Phylogenetic Analysis

#### 3.1 Align marker protein sequences using MAFFT or MUSCLE.

#### 3.2 Trim alignments with trimAl or Gblocks.

#### 3.3 Construct phylogenetic trees using:

- Maximum likelihood: IQ-TREE or RAxML with model selection.
- Bootstrap support ( $\geq 1000$  replicates) or SH-aLRT.

#### 3.4 Visualize trees with iTOL or FigTree.

#### 3.5 Root trees using appropriate outgroups.

## Phase 4: Genome Comparison and Taxonomy

- 4.1 Compute pairwise amino acid identity (AAI) between all genomes.
- 4.2 Compute gene content similarity (shared orthologous groups).
- 4.3 Apply taxonomic thresholds proposed in the paper.
- 4.4 Generate genome synteny plots to compare gene order.
- 4.5 Distinguish virophages from PLVs based on the established criteria.

## Phase 5: Validation

- 5.1 Reproduce phylogenetic trees and verify topology.
- 5.2 Reproduce AAI distribution plots and taxonomic assignments.
- 5.3 Reproduce gene content comparison tables.

## Datasets

| Dataset / Source               | Type                      | Location / Notes                                                                                                                          |
|--------------------------------|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| NCBI GenBank virophage genomes | Genome sequences          | <a href="https://www.ncbi.nlm.nih.gov/genbank/">https://www.ncbi.nlm.nih.gov/genbank/</a><br>Accession numbers from paper's supplementary |
| IMG/VR database                | Metagenomic viral genomes | <a href="https://img.jgi.doe.gov/vr/">https://img.jgi.doe.gov/vr/</a>                                                                     |
| Pfam / InterPro HMM profiles   | Protein family models     | <a href="https://www.ebi.ac.uk/interpro/">https://www.ebi.ac.uk/interpro/</a>                                                             |
| Custom virophage HMM profiles  | From paper                | Supplementary materials or author correspondence                                                                                          |
| Paper supplementary tables     | Genome lists, accessions  | Accompanying the paper                                                                                                                    |

## Models, Tools, and Software

| Tool / Library | Version    | Purpose / URL                                                                                                                    |
|----------------|------------|----------------------------------------------------------------------------------------------------------------------------------|
| Prodigal       | $\geq 2.6$ | Gene prediction<br><a href="https://github.com/hyattprodigal/Prodigal">https://github.com/hyattprodigal/Prodigal</a>             |
| HMMER          | $\geq 3.3$ | HMM-based protein search<br><a href="http://hmmer.org/">http://hmmer.org/</a>                                                    |
| MAFFT          | $\geq 7.4$ | Multiple sequence alignment<br><a href="https://mafft.cbrc.jp/alignment/software/">https://mafft.cbrc.jp/alignment/software/</a> |

| Tool / Library     | Version     | Purpose / URL                                                                                                          |
|--------------------|-------------|------------------------------------------------------------------------------------------------------------------------|
| trimAl             | $\geq 1.4$  | Alignment trimming<br><a href="http://trimal.cgenomics.org/">http://trimal.cgenomics.org/</a>                          |
| IQ-TREE            | $\geq 2.1$  | Maximum likelihood phylogenetics<br><a href="http://www.iqtree.org/">http://www.iqtree.org/</a>                        |
| RAxML-NG           | latest      | Alternative ML tree builder<br><a href="https://github.com/amkozlov/raxml-ng">https://github.com/amkozlov/raxml-ng</a> |
| BLAST+             | $\geq 2.12$ | Sequence similarity search<br><a href="https://blast.ncbi.nlm.nih.gov/">https://blast.ncbi.nlm.nih.gov/</a>            |
| ETE3               | latest      | Tree manipulation and visualization (Python)<br><a href="http://etetoolkit.org/">http://etetoolkit.org/</a>            |
| iTOL               | web         | Interactive tree visualization<br><a href="https://itol.embl.de/">https://itol.embl.de/</a>                            |
| CompareM / pyani   | latest      | AAI and ANI computation<br><a href="https://github.com/dparks1134/CompareM">https://github.com/dparks1134/CompareM</a> |
| Python (BioPython) | $\geq 3.8$  | Sequence manipulation<br><a href="https://biopython.org/">https://biopython.org/</a>                                   |

## Hardware Requirements

- Standard workstation; phylogenetic analyses can be parallelized.
- $\geq 16$  GB RAM;  $\geq 50$  GB storage for genomes and databases.